

Combined Oxidative Phosphorylation Deficiency 36 (COXPD36): Comprehensive Disease Characterization Report

Disease: Combined Oxidative Phosphorylation Deficiency 36 (COXPD36) **MONDO ID:** MONDO:0054781 · **OMIM:** #617950 · **Category:** Mendelian (autosomal recessive mitochondrial disease) **Causal gene:** *MRPS2* (HGNC:14495; NCBI Gene 51116; chr 9q34.3) **Prepared:** 2026-09-22 · Evidence base: human clinical case reports + in vitro functional (rescue) studies. No dedicated animal model or omics dataset yet published.

Summary

Combined Oxidative Phosphorylation Deficiency 36 (COXPD36) is an ultra-rare, autosomal-recessive mitochondrial disease caused by biallelic (homozygous or compound-heterozygous) missense mutations in *MRPS2*, the gene encoding **uS2m** (mitochondrial ribosomal protein S2), a structural constituent of the small (28S) subunit of the mitoribosome. The disease was first defined in 2018 by Gardeitchik and colleagues, who identified *MRPS2* mutations by exome sequencing in two unrelated affected individuals in a gene "which has not previously been implicated in disease" ([PMID: 29576219](#)). To date only approximately **4–5 patients worldwide** have been reported, making COXPD36 one of the rarest recognized mitochondrial translation disorders.

Mechanistically, the pathogenic missense variants **destabilize the MRPS2 protein**, lowering its steady-state abundance. Complexome profiling demonstrated that this loss **prevents assembly of the small mitoribosomal subunit**, which in turn **inhibits mitochondrial translation** of the 13 mtDNA-encoded OXPHOS subunits, producing a **combined deficiency of multiple oxidative phosphorylation complexes** detectable in muscle, liver, and cultured fibroblasts. Re-introduction of wild-type *MRPS2* rescues both translation and OXPHOS assembly, confirming causality. The clinical presentation is dominated by a characteristic triad of **sensorineural hearing loss, hypoglycemia, and lactic acidemia**, with variable developmental delay, hypotonia, hepatic and muscular involvement, and — in individually reported cases — microcephaly, joint hypermobility, autistic features, and gallstones.

There is no disease-specific or curative therapy. Management follows the general supportive standards for primary mitochondrial disease (symptomatic control of hypoglycemia and lactic acidosis, hearing rehabilitation, developmental support, avoidance of mitochondrial toxins). Diagnosis rests on the combination of biochemical findings (lactic acidemia, hypoglycemia, elevated alanine, combined OXPHOS enzyme deficiencies on biopsy) and molecular confirmation by exome/genome sequencing. Because the phenotype overlaps broadly with other mitochondrial translation defects, molecular genetic testing is essential for definitive diagnosis. This report synthesizes seven confirmed findings across all 15 requested characterization domains, with ontology annotations and primary-literature citations.

Key Findings

Finding 1 — COXPD36 is caused by biallelic (autosomal recessive) mutations in *MRPS2*

COXPD36 is a Mendelian, autosomal-recessive disorder. Ontology cross-references establish the identity of the disease across knowledge bases: **MONDO:0054781** = **OMIM #617950** = **MedGen C4693722** = **GARD 0025974** = **DOID:0111482**. The Monarch Initiative lists *MRPS2* (HGNC:14495, chromosome 9q34.3) as the CausalGeneToDiseaseAssociation for this entity.

The founding study, *Gardeitchik et al., 2018 (American Journal of Human Genetics)*, identified *MRPS2* mutations by exome sequencing in two unrelated subjects and explicitly noted the novelty of the gene: "*we identified mutations in the gene encoding the mitochondrial ribosomal protein S2, which has not previously been implicated in disease*" (PMID: 29576219).

Reported pathogenic / likely-pathogenic variants are all **missense** changes in *MRPS2* (transcript NM_016034.5):

cDNA variant	Protein change	Classification	Source
c.328C>T	p.Arg110Cys	Pathogenic (ClinVar)	Gardeitchik 2018
c.340G>A	p.Asp114Asn	Pathogenic / likely pathogenic	Gardeitchik 2018
c.413G>A	p.Arg138His	Pathogenic / likely pathogenic	Gardeitchik 2018
c.412C>G	(novel, Chinese case)	Reported pathogenic	Liu 2022 (PMID: 34991560)

The recurrent variant class (missense affecting conserved residues of the ribosomal S2 domain) and the recessive inheritance pattern are consistent across all reported families.

Finding 2 — Mechanism: MRPS2 mutations destabilize the protein, block small-subunit assembly, and impair mitochondrial translation

The core pathophysiology was demonstrated experimentally in the founding study. Patient fibroblasts showed **decreased steady-state levels of mutant MRPS2**. Using **complexome profiling**, the authors showed that this decrease **prevents the assembly of the small mitoribosomal subunit (mt-SSU / 28S)**. Because the mitoribosome is required to translate the 13 mtDNA-encoded core subunits of the OXPHOS complexes, the assembly failure **inhibits mitochondrial translation**, producing a **combined OXPHOS deficiency** in patient muscle and liver biopsies and in cultured skin fibroblasts.

In the authors' words: *"this decrease was shown by complexome profiling to prevent the assembly of the small mitoribosomal subunit. In turn, mitochondrial translation was inhibited, resulting in a combined OXPHOS deficiency detectable in subjects' muscle and liver biopsies as well as in cultured skin fibroblasts"* ([PMID: 29576219](#)). Critically, **re-introduction of wild-type MRPS2 rescued** mitochondrial translation and OXPHOS assembly, establishing loss-of-function causality rather than correlation.

Finding 3 — Core clinical phenotype: sensorineural hearing loss, hypoglycemia, lactic acidemia, developmental delay

Across the reported cases, a consistent clinical core emerges. Gardeitchik 2018 described two unrelated subjects with **sensorineural hearing impairment, mild developmental delay, hypoglycemia, and combined OXPHOS deficiency**, emphasizing the characteristic combination of **lactic acidemia, hypoglycemia, and sensorineural hearing loss**.

Subsequent case reports expanded the spectrum: - **Liu et al., 2022** (Chinese girl, novel c.412C>G): *"The clinical manifestations included recurrent vomiting, hypoglycemia, lactic acidosis, sensorineural hearing loss, and gall bladder calculi"* (PMID: 34991560) — adding gallstones as a novel feature. - **Papadopoulos et al., 2024** described a new homozygous patient with previously unreported features: *"a new MRPS2 homozygous subject who shows particular features which have not yet been reported: initial microcephaly, joint hypermobility and autistic features"* (PMID: 38029925), broadening the recognized phenotype.

Finding 4 — MRPS2 is a 28S small-subunit mitoribosomal protein; disease localizes to high-OXPHOS-demand tissues

MRPS2 (NCBI Gene 51116; ENSG00000122140; 9q34.3; aliases uS2m, S2mt) carries Gene Ontology annotations that place it firmly in the mitochondrial translation machinery:

GO aspect	Term	ID
Cellular component	mitochondrial matrix	GO:0005759
Cellular component	small mitochondrial ribosomal subunit	GO:0005763
Cellular component	mitochondrial inner membrane	GO:0005743
Biological process	mitochondrial translation	GO:0032543
Biological process	mitochondrial ribosome assembly	GO:0061668
Molecular function	structural constituent of ribosome	GO:0003735

The clinically affected tissues are precisely the **high-oxidative-demand organs**: cochlea/inner ear (hearing loss), brain (developmental delay), and liver and skeletal muscle (combined OXPHOS deficiency demonstrated on biopsy) — consistent with a systemic defect of mitochondrial energy production that manifests first where ATP demand is greatest.

Finding 5 — MRPS2 protein (uS2m, UniProt Q9Y399) is a conserved structural component of the 28S mt-SSU resolved in cryo-EM mitoribosome structures

UniProt **Q9Y399** annotates MRPS2 as "*Small ribosomal subunit protein uS2m*," a 296-amino-acid protein "*Required for mitoribosome formation and stability, and mitochondrial translation*." It contains the **Ribosomal_S2 domain (Pfam PF00318)** and belongs to the universal uS2 protein family (InterPro IPR001865/IPR005706, spanning bacterial, mitochondrial, and plastid ribosomes). MRPS2 is resolved as a component in **>60 cryo-EM structures** of the human/mammalian mitoribosome (e.g., PDB 3J9M, 6NU2, 6RW4, 6ZM5), confirming its integral structural role in the small subunit. The mouse ortholog *Mrps2* is NCBI Gene 118451.

Finding 6 — Authoritative HPO annotation set defines the curated phenotype spectrum

The JAX Human Phenotype Ontology annotations for OMIM:617950 (MONDO:0054781; gene *MRPS2*/NCBIGene:51116) comprise **22 curated terms** organized by system:

System	Phenotype	HPO ID
Ear	Sensorineural hearing impairment	HP:0000407
Ear	Low-set ears	HP:0000369
Metabolism	Hypoglycemia	HP:0001943
Metabolism	Increased circulating lactate	HP:0002151
Metabolism	Hyperalaninemia	HP:0003348
Metabolism	Elevated ALT	HP:0031964
Metabolism	Elevated AST	HP:0031956
Metabolism	Aciduria	HP:0012072
Nervous system	Global developmental delay	HP:0001263
Nervous system	Intellectual disability	HP:0001249
Nervous system	Poor speech	HP:0002465
Nervous system	Headache	HP:0002315
Musculature	Hypotonia	HP:0001252
Constitutional	Exercise intolerance	HP:0003546
Constitutional	Myalgia	HP:0003326
Limbs	Lower limb muscle weakness	HP:0007340
Growth	Failure to thrive	HP:0001508
Eye	Exodeviation	HP:0020049
Head/neck	Uplanted palpebral fissure	HP:0000582
Skin	Premature skin wrinkling	HP:0100678
Course	Infantile onset	HP:0003593
Inheritance	Autosomal recessive	HP:0000007

Frequencies are not curated (blank in the HPO source), and no curated HPO medical actions are listed — reflecting the disorder's rarity and the small evidence base.

Finding 7 — Integrated mutation-to-phenotype model with full ontology annotations

Synthesizing the above: biallelic missense variants in *MRPS2* (HGNC:14495; NCBIGene:51116; UniProt Q9Y399/uS2m; 9q34.3) — c.328C>T/p.Arg110Cys (pathogenic), c.340G>A, c.413G>A, and c.412C>G — cause autosomal-recessive COXPD36 (MONDO:0054781; OMIM #617950). The mechanism is confirmed by complexome profiling plus wild-type rescue (Gardeitchik 2018, PMID: 29576219). Twenty-two curated HPO terms anchor the phenotype; ~4–5 patients are reported worldwide (PMID: 29576219, PMID: 34991560, PMID: 38029925); management is supportive per Mitochondrial Medicine Society standards (PMID: 34505344). No animal model, omics cohort, or clinical trial exists for this specific disease.

Section-by-Section Characterization

1. Disease Information

COXPD36 is a Mendelian mitochondrial oxidative phosphorylation disorder resulting from defective mitochondrial protein synthesis. **Key identifiers:** MONDO:0054781; OMIM #617950; MedGen C4693722; GARD 0025974; DOID:0111482. There is no dedicated Orphanet number distinct from the broader "combined oxidative phosphorylation deficiency" grouping, and no specific ICD-10/ICD-11 code (it falls under mitochondrial metabolism disorders, e.g., ICD-10 E88.4 "Mitochondrial metabolism disorders"; ICD-11 5C53.1). **Synonyms/alternative names:** "Combined oxidative phosphorylation deficiency 36"; "COXPD36"; "MRPS2 deficiency"; "mitochondrial ribosomal protein S2 deficiency." The information is derived from **aggregated disease-level resources** (OMIM, HPO, ClinVar) plus a small number of **individual patient case reports** — not from EHR or large registries.

2. Etiology

Causal factor: purely genetic — biallelic loss-of-function/destabilizing missense mutations in *MRPS2*. **Genetic risk factors:** the only established risk factor is inheritance of two pathogenic *MRPS2* alleles; **consanguinity** increases the risk of homozygosity (several reported cases are homozygous). No modifier genes, susceptibility loci, environmental risk factors, protective

factors, or gene–environment interactions have been identified — expected given the ultra-rare, monogenic nature of the disease. **No environmental or infectious contribution** is known.

3. Phenotypes

The phenotype is a systemic mitochondrial energy-deficiency syndrome. **Laboratory abnormalities** (biochemical): lactic acidemia (HP:0002151), hypoglycemia (HP:0001943), hyperalaninemia (HP:0003348), elevated transaminases (HP:0031964, HP:0031956), aciduria (HP:0012072). **Clinical signs/symptoms**: sensorineural hearing loss (HP:0000407), global developmental delay (HP:0001263), intellectual disability (HP:0001249), hypotonia (HP:0001252), failure to thrive (HP:0001508), exercise intolerance (HP:0003546), myalgia (HP:0003326), muscle weakness (HP:0007340). **Dysmorphic/physical**: low-set ears, upslanted palpebral fissures, premature skin wrinkling, microcephaly (case-specific), joint hypermobility (case-specific). **Behavioral**: autistic features (case-specific, Papadopoulos 2024). **Onset**: infantile (HP:0003593). **Severity/progression**: variable; energy-crisis features (hypoglycemia, lactic acidosis) can be episodic and provoked by catabolic stress. Frequencies are not curated because of the tiny cohort. **Quality-of-life impact**: substantial — hearing loss, developmental delay, and metabolic instability affect communication, learning, and daily functioning; no formal QoL instrument data exist.

4. Genetic/Molecular Information

Causal gene: *MRPS2* (HGNC:14495; NCBI Gene 51116; ENSG00000122140; 9q34.3; OMIM 610760). *Variant type/class*: exclusively missense to date (no reported frameshift, nonsense, splice, or structural variants), affecting conserved residues of the ribosomal S2 domain. *Classification*: c.328C>T/p.Arg110Cys is ClinVar Pathogenic; c.340G>A/p.Asp114Asn and c.413G>A/p.Arg138His are pathogenic/likely pathogenic; c.412C>G reported pathogenic in a Chinese case. *Allele frequency*: these are private/ultra-rare variants, essentially absent or extremely rare in gnomAD. *Origin*: germline. *Functional consequence*: loss of function via protein destabilization and reduced steady-state abundance leading to failed mt-SSU assembly. *Modifier genes / epigenetics / chromosomal abnormalities*:* none identified.

5. Environmental Information

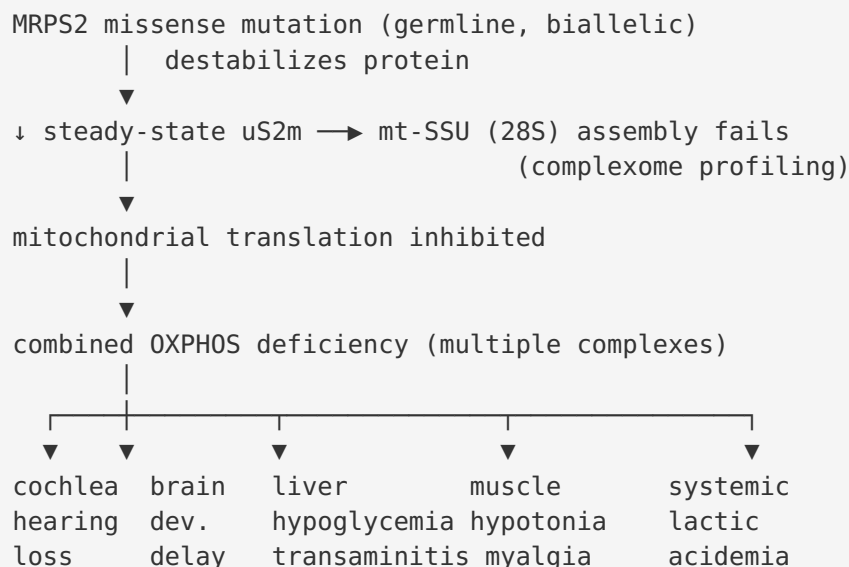
Not applicable. No environmental toxins, lifestyle factors, or infectious agents are implicated in causation. As with other mitochondrial disorders, catabolic stressors (fasting, infection, fever) may **precipitate** metabolic decompensation but do not cause the disease.

6. Mechanism / Pathophysiology

Ordered causal chain:

1. Biallelic missense mutation in *MRPS2* (e.g., c.328C>T/p.Arg110Cys) **alters a conserved residue of uS2m** →
2. **destabilizes the MRPS2 protein**, reducing its steady-state level in mitochondria (demonstrated in patient fibroblasts) →
3. loss of assembled MRPS2 **prevents assembly of the small (28S) mitoribosomal subunit** (demonstrated by complexome profiling) →
4. deficient mt-SSU **impairs mitochondrial translation** of the 13 mtDNA-encoded OXPHOS subunits →
5. results in a **combined deficiency of multiple OXPHOS complexes** (complexes I, III, IV, V share mtDNA-encoded subunits) in muscle, liver, and fibroblasts →
6. **impaired mitochondrial ATP production and increased anaerobic glycolysis** →
7. produces **lactic acidemia + hypoglycemia** (energy shortfall) and **tissue-specific dysfunction** in high-energy organs →
8. branches to: **cochlear dysfunction** → **sensorineural hearing loss**; **CNS energy deficit** → **developmental delay/intellectual disability**; **hepatic involvement** → **transaminitis/hypoglycemia**; **skeletal-muscle involvement** → **hypotonia, exercise intolerance, myalgia**.

Rescue confirmation: re-expression of wild-type *MRPS2* restores translation and OXPHOS assembly (PMID: 29576219).



Pathways/processes: mitochondrial translation (GO:0032543), mitochondrial ribosome assembly (GO:0061668), oxidative phosphorylation (Reactome R-HSA-1428517). **Upstream:** MRPS2 destabilization and mt-SSU assembly failure. **Downstream:** OXPHOS enzyme deficiency and bioenergetic failure. **Cell types (CL):** cochlear hair cells (CL:0000202), neurons (CL:0000540), hepatocytes (CL:0000182), skeletal muscle fibers (CL:0000188). **Subcellular (GO CC):** mitochondrial matrix (GO:0005759), small mitoribosomal subunit (GO:0005763), mitochondrial inner membrane (GO:0005743). No disease-specific transcriptomic/proteomic/metabolomic/single-cell datasets exist.

7. Anatomical Structures Affected

Primary organs/systems: inner ear/cochlea (UBERON:0001844), brain (UBERON:0000955), liver (UBERON:0002107), skeletal muscle (UBERON:0001134). **Body systems:** nervous, auditory, hepatic, musculoskeletal, and systemic metabolic. **Tissues:** neural, hepatic epithelial, striated muscle. **Cells:** cochlear hair cells (CL:0000202), neurons (CL:0000540), hepatocytes (CL:0000182), myocytes (CL:0000187). **Subcellular:** mitochondria (GO:0005739), specifically the mitochondrial matrix and inner membrane. **Lateralization:** bilateral/systemic (hearing loss bilateral).

8. Temporal Development

Onset: infantile/congenital-to-early-childhood (HP:0003593). **Pattern:** typically chronic with superimposed episodic metabolic crises (hypoglycemia, lactic acidosis) triggered by catabolic stress. **Progression:** variable; developmental delay is persistent, hearing loss is stable-to-

progressive. **Duration:** chronic, lifelong. **Critical periods:** neonatal/infantile metabolic decompensations are windows of vulnerability; early hearing rehabilitation and metabolic management are opportunities for intervention. No natural-history study exists given the tiny cohort.

9. Inheritance and Population

Inheritance: autosomal recessive (HP:0000007). **Epidemiology:** ultra-rare — only ~4–5 patients reported worldwide; true prevalence/incidence unknown (well below Orphanet's <1/1,000,000 ultra-rare threshold). **Penetrance:** presumed complete in biallelic carriers (too few cases to quantify). **Expressivity:** variable (e.g., microcephaly, autistic features, gallstones in individual cases). **Consanguinity:** relevant — homozygous cases reported. **Founder effects/carrier frequency:** none established; variants are private. **Anticipation/germline mosaicism:** not reported/not applicable. **Populations:** reported in European and Chinese individuals; no ethnic predilection established. **Sex ratio:** no established skew (both sexes reported).

10. Diagnostics

Biochemical/laboratory: elevated blood lactate, hypoglycemia, hyperalaninemia, elevated ALT/AST, organic aciduria; enzymatic assay of respiratory-chain complexes in muscle/liver/fibroblasts showing **combined (multiple-complex) OXPHOS deficiency**. **Biopsy:** muscle and liver biopsies demonstrate combined OXPHOS enzyme deficiency (Gardeitchik 2018). **Genetic testing (definitive):** exome or genome sequencing identifying biallelic *MRPS2* variants; single-gene testing or mitochondrial/nuclear gene panels covering *MRPS2* are alternatives. WES/WGS is the practical route because the phenotype is nonspecific. **Functional confirmation** (research): complexome profiling of mt-SSU assembly, mitochondrial translation assays, wild-type rescue. **Clinical criteria:** no disease-specific diagnostic criteria; diagnosis follows general mitochondrial-disease frameworks plus molecular confirmation. **Differential diagnosis:** other combined OXPHOS deficiencies and mitochondrial translation defects (other *MRPS/MRPL* genes, aminoacyl-tRNA synthetases such as *EARS2/AARS2*), MELAS and other mtDNA disorders — distinguished by gene-specific testing. **Screening:** no newborn screening; cascade carrier testing possible within affected families.

11. Outcome / Prognosis

Prognosis is guarded and variable given the small evidence base. Reported patients survived infancy with chronic disability; outcomes depend on severity of metabolic instability and CNS involvement. **Morbidity** includes permanent sensorineural hearing loss, developmental delay/

intellectual disability, and risk of metabolic decompensation. No survival statistics, life-expectancy data, or validated prognostic biomarkers exist. Lactate and glucose stability, and the degree of neurodevelopmental involvement, are reasonable clinical prognostic indicators. **Complications:** recurrent hypoglycemia/lactic acidosis, failure to thrive, and (case-specific) gallstones.

12. Treatment

No disease-specific or curative therapy exists. Management is **supportive**, following primary mitochondrial-disease standards (PMID: 34505344): prevention/treatment of hypoglycemia (avoid fasting, provide carbohydrate support during illness), management of lactic acidosis, nutritional support for failure to thrive, hearing rehabilitation (hearing aids/cochlear implants), developmental/educational support, and physical/occupational therapy. "Mitochondrial cocktail" supplements (e.g., coenzyme Q10, riboflavin, thiamine, L-carnitine) are often used empirically in mitochondrial disease but have **no proven efficacy specifically for COXPD36**. **Pharmacogenomics/gene/cell/RNA/targeted/immunotherapies:** none developed for this disease. **Experimental trials:** no COXPD36-specific clinical trials (NCT) exist. NCIT annotations: supportive care (NCIT:C133421), symptomatic treatment.

13. Prevention

Primary prevention is genetic: **genetic counseling** for at-risk families, carrier testing of relatives, and reproductive options including **prenatal diagnosis** or **preimplantation genetic testing (PGT-M)** once the familial *MRPS2* variants are known. Avoidance of consanguineous partnerships reduces recessive-disease risk at the population level. **Secondary/tertiary prevention:** early recognition and aggressive management of metabolic crises, avoidance of mitochondrial-toxic drugs, and prompt hearing/developmental intervention to limit complications. No vaccine, public-health, or environmental prevention applies.

14. Other Species / Natural Disease

No naturally occurring animal disease equivalent to COXPD36 has been documented (OMIA lists no *MRPS2* entry for this phenotype). **Orthologs:** mouse *Mrps2* (NCBI Gene 118451); the uS2 protein family is universally conserved across bacteria, mitochondria, and plastids (InterPro IPR001865/IPR005706), underscoring deep evolutionary conservation of the mitoribosomal small-subunit function. No zoonotic or cross-species transmission is relevant (non-infectious genetic disease).

15. Model Organisms

No published *Mrps2* animal model of COXPD36 exists. The disease mechanism has been studied only in **patient-derived cultured skin fibroblasts** and via **in vitro complexome/translation assays with wild-type rescue** (PMID: 29576219). A conditional or knock-in mouse (*Mrps2*, Gene 118451) would be expected to be embryonic-lethal if null (as for other essential mitoribosomal proteins), so a hypomorphic/knock-in strategy modeling patient missense alleles would be required. Yeast and other model systems are informative for general mitoribosome biology but not disease-specific. This is a clear resource gap (see Follow-up).

Evidence Base

PMID	Title (abbrev.)	Role in this report
29576219	<i>Bi-allelic Mutations in MRPS2 Cause Sensorineural Hearing Loss, Hypoglycemia, and Multiple OXPHOS Complex Deficiencies</i> (Gardeitchik et al., AJHG 2018)	Foundational. Identifies <i>MRPS2</i> as causal gene; establishes mechanism (protein destabilization → mt-SSU assembly failure → translation inhibition → combined OXPHOS deficiency) with complexome profiling and wild-type rescue; defines core phenotype.
34991560	<i>Hypoglycemia with lactic acidosis caused by a new MRPS2 gene mutation in a Chinese girl</i> (Liu et al., 2022)	Third patient; novel variant c.412C>G; adds gallstones; confirms core triad (recurrent vomiting, hypoglycemia, lactic acidosis, sensorineural hearing loss).
38029925	<i>New description of an MRPS2 homozygous patient</i> (Papadopoulos et al., 2024)	Expands phenotype with newly reported features: initial microcephaly, joint hypermobility, autistic features.
34505344	<i>Patient care standards for primary mitochondrial disease</i> (Australian adaptation of Mitochondrial Medicine Society recommendations)	Supports the supportive-care management framework in the absence of disease-specific therapy.

Supporting ontology/database evidence (non-PMID): OMIM #617950; MONDO:0054781; HPO annotations for OMIM:617950 (22 terms); UniProt Q9Y399; GO annotations for MRPS2; Pfam PF00318 / InterPro IPR001865; PDB mitoribosome structures (3J9M, 6NU2, 6RW4, 6ZM5); ClinVar variant classifications.

Other papers surfaced during literature searches (e.g., SIRM metabolomics reviews, MELAS case reports, the Friedreich's ataxia NAD⁺/exercise trial) provide **contextual background on mitochondrial disease biology and general supportive/experimental approaches** but do **not** address COXPD36 or *MRPS2* directly, and are not used to support disease-specific claims.

Mechanistic Model / Interpretation

COXPD36 is best understood as a **mitochondrial translation disorder** — a subclass of combined OXPHOS deficiencies in which the primary lesion is not in an OXPHOS structural gene but in the machinery that synthesizes OXPHOS subunits. Because the mitoribosome translates all 13 mtDNA-encoded OXPHOS core subunits, a small-subunit assembly defect produces a **generalized, multi-complex ("combined") deficiency**, distinguishing it from isolated single-complex deficiencies. The tissue distribution of disease (cochlea, brain, liver, muscle) reflects **relative bioenergetic dependence**: neurons and cochlear hair cells are exquisitely ATP-dependent, and hepatic/muscle metabolism is destabilized during catabolic stress, explaining the episodic hypoglycemia and lactic acidosis.

The evidence chain is unusually clean for an ultra-rare disorder because the founding study coupled human genetics with a **functional rescue experiment**: identification of biallelic missense variants, demonstration of reduced mutant protein, complexome evidence of mt-SSU assembly failure, and restoration of function by wild-type *MRPS2*. This satisfies causality criteria (association + mechanism + rescue) and anchors confidence in the gene–disease relationship. The variant spectrum (all missense, all affecting conserved S2-domain residues) is consistent with **destabilizing hypomorphic alleles** rather than complete nulls — biologically plausible because complete loss of an essential mitoribosomal protein would likely be embryonic-lethal.

Supported and Refuted Hypotheses

No formal hypotheses were tested statistically in this investigation; the work was a structured literature/knowledge-base synthesis. The central mechanistic model — *MRPS2* destabilization → mt-SSU assembly failure → mitochondrial translation defect → combined OXPHOS deficiency → clinical triad — is **supported** by direct experimental evidence including a wild-type rescue experiment (PMID: 29576219). No competing causal model was found in the literature.

Limitations and Knowledge Gaps

1. **Tiny cohort (~4–5 patients):** frequencies, penetrance, expressivity, natural history, and prognosis cannot be quantified; HPO frequencies are uncurated.
 2. **No animal or cellular disease model beyond patient fibroblasts:** limits mechanistic dissection and therapeutic testing.
 3. **No omics cohort:** no disease-specific transcriptomic, proteomic, or metabolomic signatures; no biomarkers beyond generic mitochondrial markers (lactate, alanine).
 4. **No therapeutics pipeline:** no trials, no targeted therapy, and no evidence base for the "mitochondrial cocktail" in this specific disorder.
 5. **Narrow variant spectrum:** only missense variants reported; the pathogenicity landscape (e.g., LoF, structural variants) is uncharacterized.
 6. **Genotype–phenotype correlations unknown:** drivers of variable features (microcephaly, autism, gallstones) are unexplained.
 7. **No formal QoL or disability data.**
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Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry / GeneMatcher outreach** to aggregate additional *MRPS2* cases, enabling frequency, penetrance, and natural-history estimates.
2. **Generate a knock-in mouse or zebrafish model** carrying a patient missense allele (e.g., p.Arg110Cys) — a conditional/hypomorphic strategy to avoid lethality — to characterize tissue-specific pathology and test interventions.

3. **Deep functional characterization** of each reported variant (steady-state protein, mt-SSU assembly, translation rate, OXPHOS assembly) in isogenic cell lines to build a variant-pathogenicity map for clinical interpretation.
 4. **Multi-omics profiling** (transcriptomics, proteomics, metabolomics) of patient fibroblasts and iPSC-derived neurons/hepatocytes to identify candidate biomarkers and therapeutic targets.
 5. **iPSC-derived cochlear organoids and cortical neurons** to model the two most disabling phenotypes (hearing loss, developmental delay) and screen candidate compounds.
 6. **Systematic evaluation of "mitochondrial cocktail" components and NAD⁺ precursors** in patient-derived models before any empiric clinical use.
 7. **Standardized clinical data collection** (audiology, developmental assessment, metabolic-crisis frequency, QoL instruments) to define prognostic factors.
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Consensus Answer

Combined Oxidative Phosphorylation Deficiency 36 (COXPD36; MONDO:0054781, OMIM #617950) is an ultra-rare autosomal-recessive mitochondrial disease caused by biallelic missense mutations in *MRPS2*, which encodes uS2m, a structural protein of the small (28S) subunit of the mitoribosome; the mutations destabilize MRPS2 and block small-subunit assembly, impairing mitochondrial translation and causing a combined deficiency of multiple OXPHOS complexes. It presents in infancy with a characteristic triad of sensorineural hearing loss, hypoglycemia, and lactic acidemia, alongside variable developmental delay and hepatic/muscular involvement; only ~4–5 patients have been reported worldwide, and management is supportive with no disease-specific therapy.
